1. Which of the following methods selects features by evaluating the impact of each on a model's accuracy?

A) Variance Threshold
B) Recursive Feature Elimination
C) Pearson Correlation
D) Mutual Information

✅Answer: B) Recursive Feature Elimination

---

2. Which unsupervised technique is commonly used to reduce dimensionality but not considered feature selection?

A) PCA
B) RFE
C) Chi-Square
D) Lasso

✅Answer: A) PCA

---

3. The main drawback of using wrapper methods for feature selection is:

A) They are inaccurate
B) They require data scaling
C) They are computationally expensive
D) They don't work on categorical data

✅Answer: C) They are computationally expensive

---

4. Which method is used in Flask to accept data from a frontend via POST JSON?

A) `request.data['input']`
B) `request.input_json()`
C) `request.form.get('input')`
D) `request.get_json()`

✅Answer: **D) `request.get_json()`**

---

5. What is the goal of feature selection in machine learning?

A) To select features with the highest variance

B) To increase the number of features
C) To remove redundant and irrelevant data
D) To increase dimensionality

✅Answer: C) To remove redundant and irrelevant data

---

6. Which technique ranks features based on how useful they are for predicting the target variable?

A) PCA
B) Feature Scaling
C) Feature Importance (from trees)
D) Normalization

✅Answer: C) Feature Importance (from trees)

---

7. The difference between dimensionality reduction and feature selection is:

A) Dimensionality reduction creates new features; feature selection selects existing ones
B) Feature selection creates new features
C) Both are the same
D) Feature selection cannot be used for model training

✅Answer: A) Dimensionality reduction creates new features; feature selection selects existing ones

---

8. In a recommendation system, what does "implicit feedback" mean?

A) Numerical product rating
B) Clicks, views, and purchase history
C) User surveys
D) Item description

✅Answer: B) Clicks, views, and purchase history

---

9. Which metric is NOT commonly used to evaluate recommendation systems?

A) Precision
B) Recall
C) RMSE
D) F1 Score

✅Answer: D) F1 Score

---

10. In content-based filtering, recommendations are based on:

A) Items liked by similar users
B) Features of items the user liked before
C) Average item popularity
D) User demographics

✅Answer: B) Features of items the user liked before

---

11. Which type of recommender system is more prone to the "popularity bias"?

A) Content-based filtering
B) Collaborative filtering
C) Knowledge-based filtering
D) Context-aware filtering

✅Answer: B) Collaborative filtering

---

12. Which of the following is a disadvantage of content-based recommendation systems?

A) Cold start for new users
B) Cold start for new items
C) Dependency on metadata
D) Lack of scalability

✅Answer: C) Dependency on metadata

---

13. What is the correct route decorator for handling GET requests in Flask?

A) `@app.get('/endpoint')`
B) `@app.route('/endpoint', methods=['GET'])`
C) `@app.url('/endpoint', method='get')`
D) `@flask.route('/endpoint', 'GET')`

✅Answer: **B) `@app.route('/endpoint', methods=['GET'])`**

---

14. In collaborative filtering, similarity between users or items is often calculated using:

A) Cosine similarity

B) Euclidean distance
C) Jaccard similarity
D) All of the above

✅Answer: D) All of the above

---

15. Which recommendation approach uses rule-based logic rather than machine learning or statistics?

A) Content-based filtering
B) Knowledge-based filtering
C) Collaborative filtering
D) Deep learning recommender

✅Answer: B) Knowledge-based filtering